

12

Cluster Analysis

CHAPTER OBJECTIVES

- Understand the essentials of how cluster analysis works.
- Why cluster analysis cannot define the nature of a cluster.
- Appreciate why seeking a cluster solution may be an elusive pursuit and how there may not be any substantive clusters present in your data.
- Why the determination of the existence of clusters is as much of a scientific decision as it is statistical and can often be subjective.
- How to conceptually and methodologically relate and distinguish cluster analysis from analysis of variance and discriminant analysis.
- How cluster analysis is based on a variety of distance measures and why selecting a distance measure is an important step in cluster analysis.
- Distinguish between k-means and hierarchical cluster analysis.
- How to interpret a dendrogram when performing hierarchical clustering.

Cluster analysis is a multivariable technique that seeks to uncover or identify **natural groupings** of objects for which on each object a number of measurements were taken (Everitt et al., 2001). In the machine learning and statistical learning fields, cluster analysis is usually housed in the domain of **unsupervised learning techniques**. “**Unsupervised**” here generally means that the user wishes to explore the data (or “allow the data to speak for itself”) without any preconceived notions regarding the number of clusters one may find. As we will see, however, techniques such as **k-means** cluster analysis often require the user to specify in advance the number of clusters he or she seeks to uncover, and hence, in a sense, the technique is often still **supervised** to some extent, especially when used in this way. As was the case for principal components analysis, however, it is the spirit of the word that matters. That is, if you are doing cluster analysis, you are in a strong sense trying to see what the data have to say without imposing too much of your own a priori theoretical constraints. If you are doing **discriminant analysis** or analysis of variance (ANOVA) or multivariate analysis of variance (MANOVA), on the other hand, you presumably already have a grouping structure in mind. ANOVA, MANOVA, discriminant analysis, and similar techniques? **Supervised**. Principal components, factor analysis, and cluster analysis?

Unsupervised. We could write a book critically dissecting these labels and evaluate why they can be problematic and incomplete, but let's get on instead with the purpose of the chapter (i.e. the point is to use them as convenient guidelines only, not rigid ones). The distinction, in a sense, is similar to that between **exploratory** vs. **confirmatory** factor analysis. No technique is truly “exploratory.” The use of it may be exploratory or unsupervised, but the technique itself is, again, built on abstract mathematics. Always remember that the mathematics are “innocent.” It is the use of the technique that typically drives its substantive purpose and whether it is exploratory or confirmatory, unsupervised or supervised.

The variables in cluster analysis are typically **continuous** in nature, though work has also been done in developing techniques to cluster non-continuous variables such as dichotomous or polytomous ones. In this chapter, we consider only the case of continuous variables. But what is a cluster? As we will see, and strange as it may seem, cluster analysis is unable to answer this question! That's correct, cluster analysis is unable to answer the question regarding what is a cluster (Hennig, 2015; Izenman, 2008). However, it does make an attempt to define the nature of a cluster by operationalizing it based on a variety (i.e. not just **one**) of **distance** measures. That is, the very definition of a cluster depends, from a technical sense at least, on how we define it **mathematically**. Otherwise, it has no precise meaning and is simply a word. Recall our discussion earlier in the book about what “prediction” means in regression. Unless you define it precisely mathematically, it is simply a word, and if the researcher using it is not familiar with how it is being defined mathematically, then it becomes a nonsense statement. We will survey a few of these distance measures in this chapter as they are vital to understand how something called a “cluster” can even exist in the first place.

In the age of the COVID-19 pandemic, obtaining an intuitive understanding of cluster analysis is not difficult. Indeed, the study of COVID-19 across the world has largely been a study of clusters in one way or another. One might say that the job of epidemiologists (those who study infectious diseases) over the course of the COVID-19 pandemic has been one of identifying **clusters of disease** and then hypothesizing what may be the commonality uniting one cluster vs. another. As an example, consider the COVID-19 map of Montana by county, as of October 2020 (Figure 12.1).

For these data, the clusters are predetermined and defined by county. The total number of cases (in terms of a shade) is given within each county. An epidemiologist looking at this map may ask why some counties have more cases in them than others. From this perspective, the counties can be considered as clusters. Perhaps density of population is a contributing variable or maybe popularity of mask usage is another, or maybe it is the number of individuals being tested that is the dominant variable, or the accessibility to testing services. The list goes on and on regarding what could be the potential reasons for the number of cases in some counties versus others. Note carefully that for these data, the variable of “county” is a **categorical variable**, akin to what we would have in ANOVA as an independent variable. We will return to this point later.

Now, to truly understand cluster analysis, imagine for a moment that we remove the county lines in the map. That is, for a moment pretend they do not exist. Erase them in your mind such that all you see are the different shades of density across the map. Under these circumstances, we might still ask a similar question as before, but this time, assume that as of yet we have no awareness of what might be the clustering variable. That is, we ask:

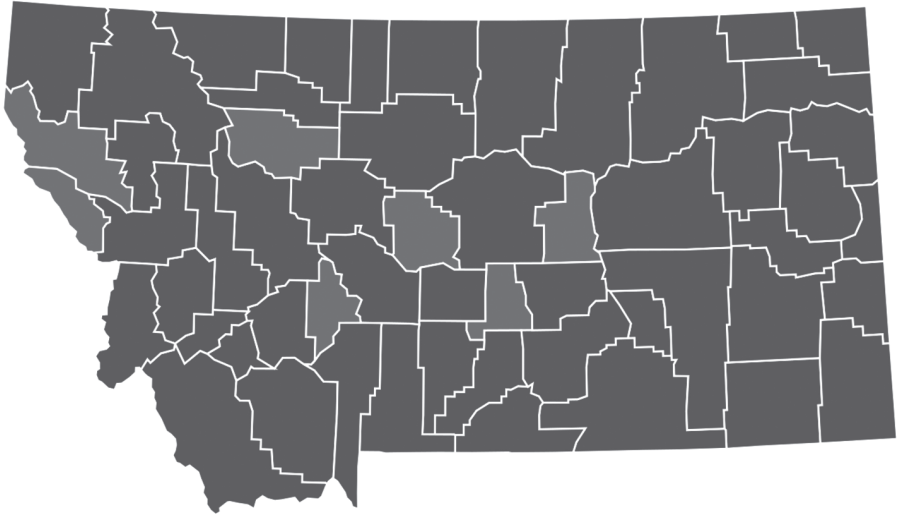
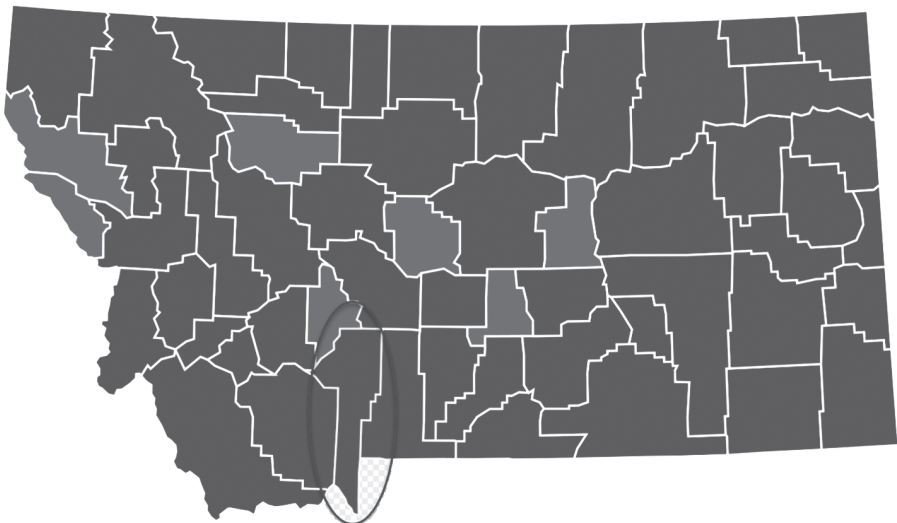


Figure 12.1 COVID-19 Map of Montana Counties during the Pandemic of 2020.

What is responsible for the presence of apparent clusters?

Why do some areas contain more cases than others? Now if we then analyzed the data further and gathered membership information on cases, we might begin to theorize **county membership** as a cluster membership variable. Notice how we have worked backwards here, where now we are pretending as though we never knew the cluster membership variable **a priori**. In the absence of the cluster variable, our job would be to identify or “uncover” (in a sense) the potential underlying variable that is somehow “responsible” for the grouping of COVID-19 cases. Our theory, of course, might be county-level membership.

However, we are just getting started. Even if we correctly identified county as a **potential** cluster variable, it in no way suggests that it is the **only** cluster variable or even the most important. For example, consider the county of **Gallatin county**, circled in the map:



Assuming we identify this as a cluster, the question from a scientific point of view is why does this county contain such density? Individuals share the same county, yes, but that does not **explain** why more cases are in that county versus others. We merely know by some measure **quantifying proximity** that these cases are more “similar” within the county than between counties. Again, notice the parallel to ANOVA here, as the idea of “within” vs. “between” variation is central to cluster analysis as it is to ANOVA.

Back to our explanation as to why we might be observing a cluster for Gallatin. We might hypothesize that the number of COVID-19 cases is high because of the ease of transmission of the disease. Hence, imagine if the epidemiologist stopped there. That is, suppose we stopped on this explanation. Attractive as the cluster explanation may be, it may still not be getting at the root or causative factor behind the cluster membership. Here are some potential alternative explanations:

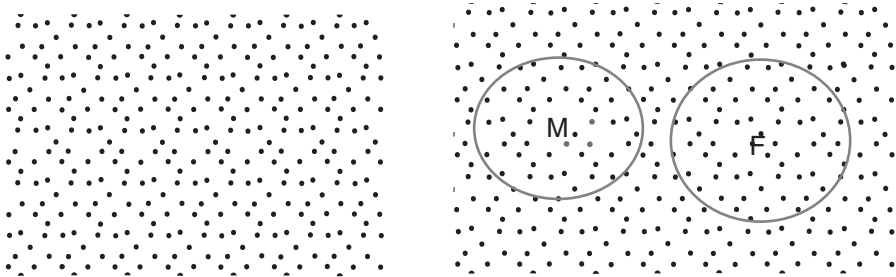
- The county may have low population density, but has had numerous close gatherings at the time COVID was in full force, and hence it encouraged ease of transmission.
- The county may feature individuals who travelled out of the county frequently, potentially contracting the disease from neighboring counties and bringing it “home” to their county.
- The county may be one that collectively resisted wearing face masks or coverings during the outbreak, contributing to ease of transmission.
- The county may consist mainly of elderly individuals, which evidence presumably showed at the time of this writing more easily contracted the disease compared to younger persons.
- The county may feature individuals with pre-existing conditions, possibly making the contraction of the disease easier.

Again, the theories the epidemiologist may advance are endless. The point of this discussion here is to emphasize that even if clusters are found and identified, learning **why** the cluster membership exists on a substantive level can be tremendously difficult, and, in some cases, virtually impossible. As we will see, cluster analysis itself is not able to answer the most important question of cluster analysis. The best cluster analysis can do is identify clusters **numerically**. It cannot identify them **substantively**. What this means is that cluster analysis can only tell you of possible groups that may exist in your data, but cannot on its own tell you why these groups might exist or their underlying **etiology**. Only the scientist can do that. Sounds limiting, doesn’t it? For the case of COVID-19, we are likely to arrive at an answer. However, for unknown data with no hypothesized cluster structure, the pursuit may prove elusive and the ethical scientist should be aware of this possibility upfront and report the absence of a solution if that is what the algorithm, combined with the science, suggests. Cluster analysis is an elegant mathematical procedure, but substantively is often a “shot in the dark.” Sometimes it works out, other times, it does not.

12.1 Cluster Analysis vs. ANOVA vs. Discriminant Analysis

As discussed throughout the book, many if not most statistical methods have technical similarities. Having surveyed ANOVA and discriminant analysis already, we are in a good position to compare them, at least on a methodological level, to the cluster

analysis framework of the current chapter. Notice that in cluster analysis, **we are unaware of group membership**. This is the exact reason why we are performing the cluster analysis. In contrast, in ANOVA, MANOVA, and discriminant analysis, group membership has already been defined on one or more variables. Referring to the case of COVID once more, suppose that based on incoming data we began to hypothesize that COVID rates were higher among men than women. This becomes our grouping variable. Framed in the context of cluster analysis, we have two clusters of observations, one of males and one of females. To visualize this, consider the following scatter of cases:



In this example, once we identify the males vs. females (M vs. F), two clusters appear to emerge (right). Hence, it may suggest that the rate of acquiring COVID-19 is different for males than it is for females. Now, had we hypothesized this might be the case beforehand and wanted to test our theory, then gender (male vs. female) would have made a suitable **independent variable**. Notice that perceived from this vantage point, cluster analysis is a more “primitive” technique from a scientific point of view (but only from a scientific point of view, not a technical point of view) when compared to ANOVA. In ANOVA, the researcher presumably has a theory as to why groups may be different, which is expressed as a factor with levels on the independent variable. How about a discriminant analysis? Since a discriminant analysis is essentially an ANOVA or MANOVA flipped around, group membership is now a response variable because we are attempting to predict the classification into one group vs. the other. Now, this is not meant to imply the procedures all have an identical technical base or are derived in exactly the same way, but it is to suggest they have similar methodological underpinnings. If you perceive these techniques from the vantage point of which to select for a given purpose, these distinctions and similarities can be helpful. ANOVA or discriminant analysis are not used as much in the spirit of “shooting in the dark” as is cluster analysis.

12.2 How Cluster Analysis Defines “Proximity”

We have said that cluster analysis seeks to generate groupings for which cases are most **similar**. How does it go about this? It is obvious that to liken observations by a measure of **proximity**, this measure must be well-defined and precise. That is, we have to mathematically define it. It turns out there are several ways of measuring proximity. These measures are often presented as measures of **distance**, or how closely or

separated two or more objects are from one another. However, whichever you choose to call them makes little difference. Though there are many numerous measures, we consider only a few of them here. Our first is by far the most popular, that of Euclidean distance.

12.2.1 Euclidean Distance

The **Euclidean distance** between two p -dimensional objects, $\mathbf{x} = [x_1, x_2, \dots, x_p]$ and $\mathbf{y} = [y_1, y_2, \dots, y_p]$ is given by

$$\begin{aligned} d(\mathbf{x}, \mathbf{y}) &= \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_p - y_p)^2} \\ &= \sqrt{(\mathbf{x} - \mathbf{y})'(\mathbf{x} - \mathbf{y})} \end{aligned}$$

where $d(\mathbf{x}, \mathbf{y})$ is the distance between the two vectors, $\mathbf{x}$ and $\mathbf{y}$. When “unpacked,” this translates to what we see on the right-hand side of the equation, the corresponding distances between observations $(x_1 - y_1)^2, (x_2 - y_2)^2, \dots, (x_p - y_p)^2$. We can easily illustrate Euclidean distance on two vectors $\mathbf{y}$ and $\mathbf{x}$ with a simple example. Consider two vectors, $\mathbf{x} = [2, 4]$ and $\mathbf{y} = [4, 7]$. The Euclidean distance between the two vectors is computed as follows:

$$\begin{aligned} d(\mathbf{x}, \mathbf{y}) &= \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_p - y_p)^2} \\ &= \sqrt{(2 - 4)^2 + (4 - 7)^2} \\ &= 3.61 \end{aligned}$$

Euclidean distances, or their squares (i.e. squared Euclidean distances) are quite popular in cluster analysis. However, other distances can be defined as well. For example, the **Minkowski metric** is given by

$$d(\mathbf{x}, \mathbf{y}) = \left[\sum_{i=1}^p |x_i - y_i|^m \right]^{1/m}$$

We can see that m is a variable that operates in the exponent. When $m = 2$, the Minkowski metric reduces to a simple Euclidean distance, as can be easily seen:

$$\begin{aligned} d(\mathbf{x}, \mathbf{y}) &= \left[\sum_{i=1}^p |x_i - y_i|^m \right]^{1/m} \\ &= \left[\sum_{i=1}^p |x_i - y_i|^2 \right]^{1/2} \\ &= \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_p - y_p)^2} \end{aligned}$$

A third measure is the **city-block distance**, also known as the **Manhattan distance**, defined by

$$d(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^p |x_i - y_i|$$

In the above, $|x_i - y_i|$ is the **absolute distance** between observations. As mentioned, the choice of which distance to use is often subjective and different distance choices may yield different cluster solutions. Often, though not always, if a cluster solution is strong in a data set, then a variety of distance measure selections should yield more or less a consistency in the cluster discovery. There is absolutely nothing wrong with a researcher trying out different distance metrics to learn what cluster solutions they provide. Such is not “cheating,” it is simply operationalizing the measure of proximity differently to see what comes about. This is the exploratory nature of cluster analysis, and trying out different distance measures is a good idea. There is no type I error rate to worry about by trying different distances in the search and potential establishment of a cluster solution. But if you find evidence for a cluster solution using some distances but not others, then that may require some explanation or discussion, which may not prove to be easy. It may require some further digging into your data to figure out why this difference is occurring. Is one distance measure emphasizing absolute distances whereas the other is not? Are there outliers in your solution that are disproportionately affecting one measure but not the other? All of these are good questions when exploring solutions with cluster analysis. It is an exercise in **dating mining**, so mine your data! In this sense, all cluster solutions are rather **subjective**, even if they are based on mathematically rigorous distance measures. “Are there clusters in my data?” “Well, it depends ...” is a good place to start, even if you do find evidence for clusters. As we said at the start of this chapter, the existence of clusters is only as good as they are defined mathematically. Otherwise, from the point of view of cluster analysis, they have no existence.

12.3 K-Means Clustering Algorithm

As mentioned, there are a variety of clustering methods. In the **k-means clustering algorithm**, a researcher first designates the number of clusters he or she is seeking to find in the data (which again, does not sound very “unsupervised”). That is, from the start, the researcher has some idea of the number he or she would like to find. Remarkably, k-means usually guarantees the user will find a cluster solution. However, the specifying of how many clusters to find is entirely subjective and must be merged with scientific considerations for the solution to usually make any sense. If the researcher believes there are two clusters underlying the data, for instance, then this is the number that is set for the algorithm to find. Similar to factor analysis, the user is free to specify the number of clusters analogous to searching for a specific number of factors. As with virtually all statistical methods, **you have to start somewhere**.

While k-means cluster analysis may seem quite modern, its origins are actually quite dated. James MacQueen (1967) best described the technique in an early paper:

The main purpose of this paper is to describe a process for partitioning an N -dimensional population into k sets on the basis of a sample. The process, which is called “ k -means”, appears to give partitions which are reasonably efficient in the sense of within-class variance ... State [sic] informally, the k -means procedure consists of simply starting with k groups each of which consists of a single random point, and thereafter adding each new point to the group whose mean the

new point is nearest. After a point is added to a group, the mean of that group is adjusted in order to take account of the new point. Thus at each stage the k -means are, in fact, the means of the groups they represent (hence the term k -means).
(pp. 281, 283)

To paraphrase MacQueen, k -means cluster analysis joins or “fuses” observations that are most similar, and does so iteratively until a **minimization criterion** has been achieved. There can be no overlap in k -means, which means that if a case is in one cluster, it cannot be in another. The cluster means that are updated at each step along the way are typically called **cluster centroids**. The idea of imposing a minimization criterion on the cluster solution is an example of other techniques studied in this book where a minimization or maximization criterion is imposed. Recall again that in least-squares regression, the objective was to fit a line such that the sum of squared errors is kept to a **minimum** value. Likewise, the objective in discriminant analysis was to find a function that **maximizes** group differences on the response. The goal of principal components analysis was to derive linear combinations of observed variables with **maximum** variance. Hence, we see again that so much of statistical modeling is a problem of optimization, that of **maximizing or minimizing** some quantity subject to constraints. It is rarely, despite what the “data deluge” zeitgeist implies, simply “allowing the data to speak for themselves.” Even if very rudimentary, a mathematical structure of some type, even if minimal, is typically still imposed on virtually any data that is subjected to statistical interpretation. “What do the data say?” The correct response is, “Well, the data say this under these conditions, operations, and constraints, and possibly something else under others.” **That is the correct answer! It is usually the only answer!** The correct response is never “The data say what they say, period.” This point cannot be emphasized enough. If you do not understand this point, you will likely make faulty scientific conclusions when interpreting data. You will be failing to appreciate and recognize the proverbial microscope through which you are observing your findings. You will be able to conduct this or that analysis as you exhibit extraordinary software skills, but you will fail to appreciate that, again by way of analogy, **the microscope is every bit as important as what is being visualized under it**. Be sure you understand this point! When a researcher says, “I found evidence for clusters,” that should be the beginning of your questions into the tool used, not the end.

12.4 To Standardize or Not?

As we will see shortly when we demonstrate cluster analysis in Python, the user has the option to leave the data as raw or to **standardize** the data, meaning to transform it to **z-scores** with a mean of 0 and variance of 1 for each variable. The decision to standardize or not is not an automatic one and the user or interpreter of cluster analysis should not assume that the decision is inconsequential for the cluster analysis. So how should you decide? Typically, if the variables are not measured on the same units and have wildly different variances, standardizing is probably a good option. However, if they are on the same units with similar variances, then you may opt not to. The best option, of course, is to compare solutions with and then without standardization; then you can see for yourself how your analysis behaves under each condition. Generally,

however, the above guideline (regarding variances) can be used as a decision as to whether or not to standardize.

12.5 Cluster Analysis in Python

We demonstrate a simple k-means cluster analysis in Python. One advantage of clustering the iris data is that we are already well aware of one potential cluster solution, that of **species**. However, for our purposes, we act as though we do not know this cluster solution or that we are seeking out a different one altogether. If we were deliberately attempting to use iris features to predict species, we would be in the realm of a supervised learning technique such as discriminant analysis or logistic regression, or MANOVA. However, since we are going to “let the data speak for themselves” as it were, we are proceeding as if we do not know a good clustering variable. Let’s first import some of the libraries and tools we will need:

```
from sklearn.cluster import KMeans
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
```

Next, we import the **iris** data from sklearn datasets, after first importing the **seaborn** package:

```
import seaborn as sns
iris = sns.load_dataset("iris")
print(iris.head())
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

We only want the first four variables, however, so we select them from the dataframe (i.e. we do not want to include species in our cluster analysis, it is categorical, and does not belong):

```
df = iris[['sepal_length', 'sepal_width', 'petal_length', 'petal_
width']]
df.head(3)
```

Out[710]:

	sepal_length	sepal_width	petal_length	petal_width
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2

Now that we have the variables we want, let’s perform the k-means cluster analysis, requesting three clusters. The number of clusters is designated by **n_clusters=3** in this case. We then use **.fit()** to obtain the solution:

```
kmeans = KMeans(n_clusters=3)
kmeans.fit(df)
kmeans.predict(df)
```

Out[721]:

```
array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 2, 2, 2, 2, 1, 2, 2, 2,
       2, 2, 2, 1, 1, 2, 2, 2, 2, 1, 2, 1, 2, 1, 2, 2, 1, 1, 2, 2, 2, 2,
       2, 1, 2, 2, 2, 2, 1, 2, 2, 2, 1, 2, 2, 2, 1, 2, 2, 1])
```

Using `kmeans.predict()` above allows us to obtain the predicted group membership for each case. For instance, the first case was predicted into the cluster = 0, the second case as well, and so on. If you count up the numbers from left to right above, you will see that the 51st case is classified into cluster = 1, and so on for the remaining cases. Let's now obtain the centroids of the cluster analysis. We can obtain this using `cluster_centers_`:

```
centroids = kmeans.cluster_centers_
centroids
```

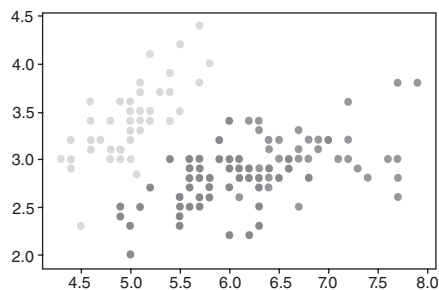
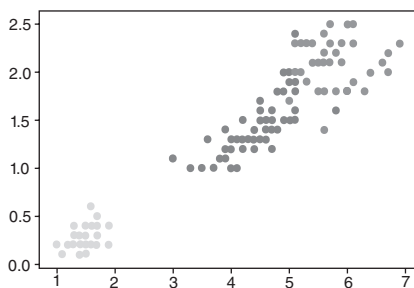
Out[727]:

```
array([[5.006,      3.428,      1.462,      0.246    ],
       [5.9016129, 2.7483871, 4.39354839, 1.43387097],
       [6.85,      3.07368421, 5.74210526, 2.07105263]])
```

We now generate a plot showing the separation achieved across petal length and petal width (left) and then across sepal length and sepal width (right):

```
kpredict = kmeans.predict(df)
plt.scatter(iris['petal_length'], iris['petal_width'], c =
kpredict, cmap = 'cool')
Out[153]: <matplotlib.collections.PathCollection at 0x1db04358>

plt.scatter(iris['sepal_length'], iris['sepal_width'], c =
kpredict, cmap = 'cool')
Out[154]: <matplotlib.collections.PathCollection at 0x21b5eac8>
```



As we can see from the plots, there appears to be pretty good separation from two of the species from that of a third, especially across petal length and petal width. How do we know species is a suitable clustering variable? Simply because we have the variable already in our data file. That is, we already have group information on our cases, **we know they are “classifiable” by species**. This is why earlier we were able to look at the first 50 cases and realize that they all belong to the species *setosa*. Had we not had the species variable, or were naïve about what a suitable clustering variable could be for these data, we would have had to do more exploratory work to discover a suitable clustering variable or give up and acknowledge that the clustering solution “makes no sense” to us as of yet. Be sure this is clear, as it is **fundamental to the nature of clustering**. It is the most important point. The clustering algorithm did not “know” species was a suitable clustering variable since we did not arm the algorithm with this information. It simply performed a cluster analysis based on the k-means algorithm. It did so “unsupervised.” It operated mathematically according to a blind algorithm. It is us, as users of cluster analysis, that “validated” the cluster solution against a variable we already knew existed in our data file, that of species. Had we not had the species variable, we would be scratching our heads wondering why those first 50 cases were classified into the first cluster, and so on for the remaining clusters. To obtain the identification case information for each cluster in the three-cluster solution, you can code the following (Larose and Larose, 2019), then compute some descriptive information on each cluster using `cluster1.describe()`, `cluster2.describe()`, etc.:

```
cluster = kmeans.labels_
cluster1 = iris.loc[cluster == 0]
cluster2 = iris.loc[cluster == 1]
cluster3 = iris.loc[cluster == 3]
```

When you try the above for the first cluster (too many numbers to list here), you will see they are all of the species “*setosa*.” And when we compute `cluster1.describe()`, we note the means featured earlier when we obtained the centroids:

```
cluster1.describe()
Out[166]:
```

	sepal_length	sepal_width	petal_length	petal_width
count	50.00000	50.000000	50.000000	50.000000
mean	5.00600	3.428000	1.462000	0.246000
std	0.35249	0.379064	0.173664	0.105386
min	4.30000	2.300000	1.000000	0.100000
25%	4.80000	3.200000	1.400000	0.200000
50%	5.00000	3.400000	1.500000	0.200000
75%	5.20000	3.675000	1.575000	0.300000
max	5.80000	4.400000	1.900000	0.600000

This is why the current cluster analysis on the iris data is “easy” from a substantive or scientific point of view, in that we already have a good “theory” as to what might be “responsible” for the clusters, that of species. In practice, clustering is rarely that easy and you may have to sit for a while, or even years (in theory) with a solution until you find a scientific explanation for it. The variable “responsible” for the clustering you are witnessing may even, in theory, be “unknowable” as of yet, and may require other

advancements in human history to actually account for it. Hence, the best clustering variable may actually be **latent** (analogous to a latent factor from the previous chapter), at least for now.

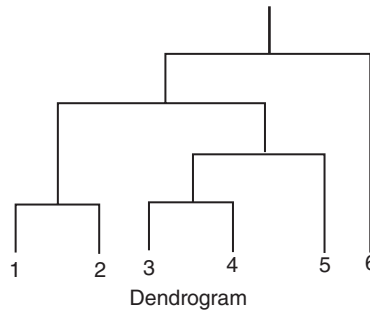
For instance, in the COVID-19 pandemic, why some people got the illness and others did not could theoretically remain unknown for a long time, perhaps even until a new gene is identified (at least in theory). Obtaining clusters is easy, those who have the illness vs. those who do not. But what differentiates the groups? That is the cluster solution you are seeking, or, in discriminant analysis terms, the response variable you are wanting to identify, or, in ANOVA terms (or *t*-test in this specific two-group situation), the levels of the independent variable you are seeking to operationalize. If you know in advance, or have a strong **theory** about what the clustering variable might be, you will have little reason to seek out a cluster solution. Instead, you will try a **discriminant analysis** or **M(ANOVA)** to evaluate your “group” or “cluster” theory. **Notice how unified, from a conceptual standpoint, all of these statistical methods really are.** That is, if you approach them from a scientific perspective, rather than a technical one, you gain a good appreciation for how these techniques are related to the theory-building interests of the scientist. This is precisely why techniques such as principal components and cluster analysis are often regarded as more “exploratory” or “unsupervised” when compared to the more “confirmatory” or “supervised” techniques of ANOVA, discriminant analysis, logistic regression, etc.

Hence, the central point here is the fact that a clustering algorithm that “found” clusters does not **necessarily** imply they exist apart from the abstract distance measures that created them. Mathematics is an abstract system and its results need not necessarily accord with our current theories or explanations of nature. This is a theme that has repeated itself throughout this book and hence is true of cluster analysis as well. Always seek to separate the mathematics from the material on which you are applying the mathematics to, at least at first. Then, as you would in putting your hands together and seeing how well you can overlay them exactly finger by finger, entertain how closely they might fit. If they fit well, you might have something of scientific value. If they do not, then you may be back to square one. Eliminating a proverbial square, however, is still progress (though sadly, often not publishable). We now survey the second primary approach to clustering, that of hierarchical or “agglomerative” clustering.

12.6 Hierarchical Clustering

In k-means clustering, we specified in advance the number of clusters we hypothesized from the data or were seeking to find. The k-means algorithm then worked iteratively to obtain a cluster solution. In a second general approach to clustering, that of **hierarchical clustering** (also often more generally going by the name of “**agglomerative**”), the researcher does not specify directly the number of clusters he or she is seeking to uncover from the data. In hierarchical clustering, the researcher allows the algorithm to parse things out and potentially uncover a cluster solution. In this sense, then, hierarchical clustering may be perceived as even more “exploratory” or “unsupervised” of a method in that it is truly exploring the data without any preconceived theory as to even the number of clusters – that is, at least no theory that the researcher is advancing overtly.

There are different methods of joining clusters in hierarchical clustering. These include **single linkage**, **complete linkage**, and **average linkage** methods. Single linkage (or “nearest-neighbor”) features merging cases with the smallest distance, then merging remaining cases with the smallest distance from the newly created cluster. The algorithm proceeds in this way at each step of the hierarchy, making successive mergers along the way. Results can be depicted in what is known as a **dendrogram**, which is a tree-like structure that reveals a history of the mergers made at each step. An example of a simple dendrogram is shown:



The dendrogram should be read from the bottom-up. We see that cases 1 and 2 were merged, as well as 3 and 4. Cases 3 and 4, forming a new cluster, were merged to case 5. The final cluster formation was between case 6 and the cluster generated by the left-hand side of the dendrogram. While in simple linkage the smallest distance is a priority for a merger, in complete linkage the largest of the dissimilarities are calculated and the smallest of these are merged at each step. In average linkage, as the name suggests, it is the average of the dissimilarities at each step that is taken as a measure of distance. For more details on these distance measures, see Johnson and Wichern (2007).

Incidentally, as recommended throughout this book, if your data is not that large, it is well worth your time to study the dendrogram linkage by linkage, fusion by fusion to better understand why two points were fused. For instance, given that cases 1 and 2 were merged, the next question is, why? How about cracking open your data file to look at those two cases? Are they similar in some sense? If they were both dogs out of all the animals you are seeking to cluster, then you may of course have a hint. In this way, good statistical analysis is encouraging you to **get to know your data better**. That’s the point! These types of merger-by-merger inspections can be very exciting on data that is well-calibrated and precise. “Why are those two animals clustering and why did they then cluster to a third? Wow, the third animal is a cat, while the other two were dogs. What’s going on?” Asking questions like this is the objective of science, and it can get pretty exciting!

So, which linkage to use? Like many decisions in cluster analysis, the decision is ultimately **subjective**. Yes, rigorous applied statistics and science can come down to subjective decisions! Those who specialize in cluster analysis often recommend trying out a few different measures to learn if they might converge on the same or similar solution. However, the issue of whether clusters truly “exist” in your data must be coupled with scientific expertise, otherwise it is simply an exercise in mathematical partitioning. Does

this sound at all familiar? It might, since we concluded something similar about ANOVA. Well, to be exact, R.A. Fisher, its founder, actually said it. Recall that he said ANOVA was a procedure of “arranging the arithmetic” and nothing more. Paraphrased, it meant that you can perform an ANOVA on any data and it will partition variability into that of **between** vs. **within**. Whether that partition makes any scientific sense for your data, however, is not a question that can be answered by ANOVA! It must be answered by your **science**. In applied statistics, as opposed to theoretical statistics, it must always be **science first, statistics second**. And so it is with cluster analysis as well. Again, cluster analysis, like many exploratory techniques, is nothing more than an attempt to partition the **signal from the noise**. And though statistical techniques are mathematically precise, their application is often quite messy. The morale is to not have unrealistic expectations of the tool. It usually cannot, on its own, answer the deeper questions you have as a scientist. Only good science can do that, by whatever statistical method is used in the pursuit. The “promise” is not with the tool. It is with the scientist.

12.7 Hierarchical Clustering in Python

We now run a hierarchical cluster analysis in Python. As when we ran k-means, we do not include the species variable. When we ran the k-means, recall that we defined a new data set without species by identifying the variables we wished to include:

```
df = iris[['sepal_length', 'sepal_width', 'petal_length',
'petal_width']]
```

We could have just as easily have dropped “species” from the iris data as follows, instead of calling on which variables we wanted to include. In the following, the data set is still iris, but it is now without species. We execute this via `iris.drop()`:

```
iris.drop(['species'], axis=1, inplace = True)
iris
```

Out[241]:

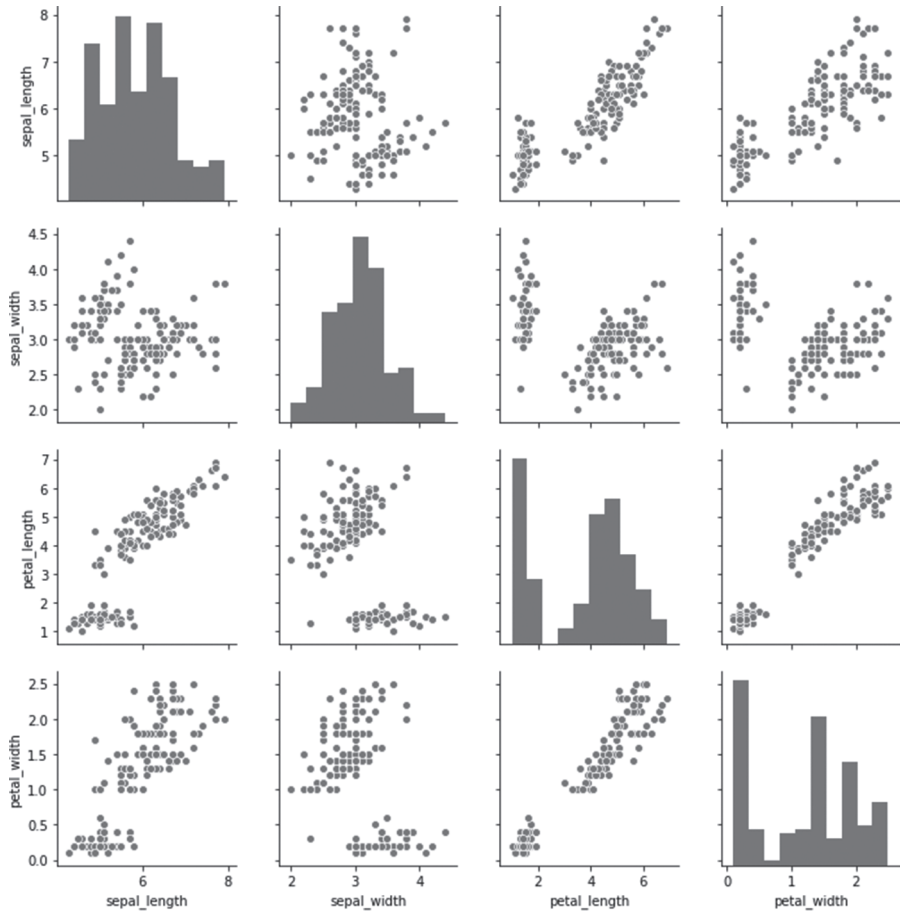
	sepal_length	sepal_width	petal_length	petal_width
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2
5	5.4	3.9	1.7	0.4

Before conducting the cluster analysis, let’s first obtain some **pairwise plots** of the sepal and petal features through **seaborn**, specifically `pairplot()`:

```
import seaborn as sns
sns.pairplot(iris)
```

Out[242]: <seaborn.axisgrid.PairGrid at 0x871e7b8>

We can see from these figures that some groups within the bivariate plots seem to differentiate themselves. For example, in the plot sepal width by petal length, notice that



the scatter is not bivariate uniform, but rather seems to contain a degree of separation. We see similar separation in the bivariate plot corresponding to petal length and petal width. Although these plots do not indicate what the final cluster solution will look like in the end, they are nonetheless informative as an exploratory portrayal of the data before conducting the analysis.

Let's now conduct the cluster analysis requesting a total of three clusters, using **Euclidean** distance with single linkage:

```
from sklearn.cluster import AgglomerativeClustering
groups = AgglomerativeClustering(n_clusters=3,
affinity='euclidean', linkage='single')
groups.fit_predict(iris)
```

By using `fit_predict()` above, as in the case of k-means earlier, we obtain the predicted cluster grouping for each case below. Because we named our cluster object “groups,” we could have also requested the cluster membership classification results using `print(groups.labels_)`. For example, the first case is predicted to go into cluster 1, the second into cluster 1, and so on. Notice that the first 50 cases (in bold) belong to the first cluster.

```
Out [502]:
array([1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0], dtype=int64)
```

Now, what are those first 50 cases? That is, what might be “responsible” or “associable” (yet not necessarily “causal”) for clustering those 50 cases? Well, let’s look at our data set to see if we can find any clues. What were the first 50 cases? The first six cases are listed (of the original iris data set, not the one in which we removed the species variable):

```
iris
Out [266]:
```

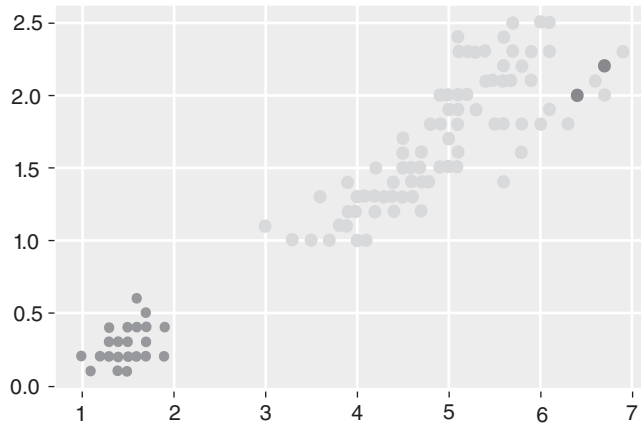
	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
5	5.4	3.9	1.7	0.4	setosa

Though we have printed only the first six cases (due to keeping things reasonable spacewise), we see that these are all of the species setosa. If you look at the rest of the list of iris data in Python, you will see the remainder of the first 50 cases are also of species setosa. Hence, as we tentatively concluded when performing the k-means analysis, one interesting clustering idea is that the first cluster is that of setosa! Monumental! Now, does that necessarily define the cluster? No. “What?” I said no! The cluster need not be defined by setosa for certain. **It need not be defined by setosa at all. True cluster membership could, in theory, be something for which setosa is merely a correlate.** Ah, correlational research, and why your instructor keeps telling you to do experiments! However, for our purposes, identifying the cluster as setosa is simply too convenient, and we will go with that. The point is, however, that defining the nature of clusters is, again, a subjective decision, and what first may appear as the defining feature may be, as mentioned, merely a **correlate to an even more definitive feature**. Always allow for this possibility when seeking an explanation for cluster membership. Just because you found a clustering solution does not mean it is “the” solution. **All models are wrong, some are useful. All scientific conclusions are tentative, some are useful.**

Next, we generate a bivariate plot of petal length and petal width to visualize the separation. We do so through `plt.scatter()`:

```
plt.scatter(iris['petal_length'], iris['petal_width'], c = groups.
labels_, cmap='cool')
```

```
Out [262]: <matplotlib.collections.PathCollection at 0x20f05a90>
```



As shown, the group *setosa* is separating itself quite well from the other two species of flower. Again, in this case, we know there are three clusters, or at least we can presume there are because of the species variable. In other cases, however, the decision may be less clear. There are criteria that can be used to identify the number of clusters in data if results are ambiguous or you have virtually no theory guiding you, but these are beyond the scope of this book. These involve, in part, plotting the total error of the cluster solution by the number of clusters for varying values of k . For details of how this is done in Python, see Grus (2019). Typically, however, an applied scientist will determine the number of clusters based on substantive insight into the problem at hand, and hence in most applied cases, “blind statistical determination” of how many clusters “exist” in the data is much too “abstract” of a solution and ignores the material that is being clustered. And as we have discussed, the “existence” of clusters is laden with philosophical difficulty, but still, such statistical metrics can aid in helping one determine a tentative solution, especially if the data are quite large and you have absolutely no theory guiding you. As we have seen with most of the techniques surveyed in this book, such decisions have to usually be merged with the science you are conducting, and not independent from it. That is, if a three-cluster solution makes more sense scientifically than a four-cluster solution, it should be the three-cluster solution that is typically adopted, at least until new insights emerge as to why the four-cluster solution may be feasible. Otherwise, as we have alluded to throughout this book, you are simply a **servant to the algorithm**, and not merging it with the material you are actually studying. The algorithm, like the computer in the Boeing 737 max (recall those two airline crashes in 2018–2019), cannot guarantee for certain the plane is flying right. The pilot needs to be able to “override” the system when required. Good scientists use statistical modeling to help them learn more about what is being modeled. Though you must be very familiar with the algorithm, your job is not to “appease” it at the cost of your science. Too often, otherwise good scientists become immersed in tweaking their data to fit the model, rather than focusing on fitting the model to the data.

Review Exercises

1. Discuss the purpose of **cluster analysis**. What might motivate a researcher to want to conduct one?
2. How is **cluster analysis** similar in some regards to **factor analysis** or **principal components analysis**? How is it different?
3. In the chapter, it was hypothesized that county might be a useful metric for measuring clusters in Montana during the COVID-19 pandemic. While it may be convenient to use county as a potential clustering variable, why might it not be the best one to explain COVID-19 clusters? What other clustering variables might be better for the state of Montana?
4. Why is the question of the **causes of cluster membership** potentially elusive? In other words, why is it such a difficult question to answer?
5. Related to exercise 4, suppose scientists found a COVID-19 cluster in the United States. Why is determining the “cause” of that particular cluster such a difficult if not impossible question to answer? Why is establishing correlates to cluster membership insufficient?
6. Differentiate between **cluster analysis** and **discriminant analysis**. How are they different? Similar?
7. Why is determining case “**proximity**” a potentially elusive pursuit?
8. Why is cluster analysis not able to answer the question “**What is a cluster?**”
9. Briefly describe the **k-means** approach to clustering. That is, briefly unpack the algorithm. How does it work, in general?
10. Give a rationale for **standardizing** one’s data before performing a cluster analysis. Is there anything wrong with comparing cluster solutions across unstandardized and standardized solutions?
11. Consider the **HCV data** from the **Machine Learning Repository**, featuring laboratory values of blood donors and Hepatitis C patients. Perform a **k-means** cluster analysis on variables ALB through to PROT. Try out solutions for two to five clusters, and comment on any findings.
12. Perform a **hierarchical** cluster analysis for the data in #11 and compare results.